But the wealth of options can seem overwhelming, and the relative ease of modifying a dataset to appear in a streamgraph, treemap, or histogram tends to promote the idea that information visualization is more about style than substance. Fortunately, well-established rules dictate what charts and methods to use for different types of data from different systems. Although we can't cover every rule in the book, we'll touch on ones that are useful to consider as we create more complicated information visualizations. Although some developers use D3 to revolutionize the use of color and layout, most simply want to create visual representations of data that support practical concerns. Because D3 is being developed in this mature information visualization environment, it contains numerous helper functions to let developers worry about interface and design rather than color and axes.

Still, to properly deploy information visualization, you should know what to do and what not to do. The best way to learn this is to review the work of established designers and information visualization practitioners, and you need to have a firm understanding not only of your data but of your audience. Although an entire library of works deals with these issues, here are a few that I've found useful and can get you oriented on the basics:

- *The Visual Display of Quantitative Information Envisioning Information*, Edward Tufte
- *Designing for Information*, Isabel Meirelles
- *Pattern Recognition*, Christian Swinehart

These are by no means the only or most applicable texts for learning data visualization, but I've found them useful for getting started. You should pare down and establish fundamental, even basic, data visualization practices that clearly represent the trends that are salient to your audience. When in doubt, simplify—it's often better to present a histogram than a streamgraph, or a hierarchical network layout (like a dendrogram) than a force-directed one. The more visually complex methods of displaying data tend to inspire more excitement, but can also lead an audience to see what they want to see or focus on the aesthetics of the graphics rather than the data.

Infoviz tip: kill your darlings

One of the best pieces of advice when it comes to working in information visualization comes from the practice of writing: "Kill your darlings." Just as writers may become enamored of certain scenes or characters, you can become enamored of a particularly elegant or sophisticated-looking graphic. Your love of a cool chart or animation can distract you from the goal of communicating the structure and patterns in the data. Remember to save your harshest criticism for your most beloved pieces, because you may find, much to your chagrin, that they're not as useful and informative as you think they are.

One thing to keep in mind while reading about data visualization is that the literature is often focused on static charts. With D3 you'll be making interactive and dynamic